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I. Aleksander W.V. Thomas P.A. Bowden

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WISARD • a radical step forward in image recognition

Professor I. Aleksander, W. V. Thomas and P. A. Bowden

The WISARD recognition system invented at Brunel University has been developed into an industrialised product by Computer Recognition Systems under licence from the British Technology Group. Using statistical pattern classification it already shows great potential in rapid sorting, and research indicates that it will track objects with positional feedback, rather like the human eye.

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WISARD is named after its inventors: WIlkie, Stoneham and Aleksander's Recognition Device. It marks a radical step forward in image processor design, offering recognition and judgement capabilities approaching that of a trained human, while maintaining the consistency of machine vision. This is achieved without the need for high-level applications programming by using statistical pattern classification techniques, implemented in hardware to give real-time performance.

The CRS WISARD is based on 15 years' of research and simulation. A prototype was built at Brunel University in 1981. Computer Recognition Systems Ltd has taken the concept under exclusive licence from the British Technology Group, and benefiting from the availability of cheap dynamic RAM memory has designed a flexible, powerful and economic industrial vision system. Combined with the company's 68000-based CRS1000 automatic inspection system, WISARD is well suited to applications in inspection, robot vision and surveillance.

The notional structure of WISARD is shown in Fig. 1. Operation may be summarised as follows. The incoming image is stored and digitised. Grey-scale images may be processed but the technique is most easily illustrated using a thresholded, binary image. In this case each pixel is represented by a single bit. A number 'n' of such bits, taken from the image using a random, but fixed mapping forms an n-tuple. A 512 by 512 binary image contains 2^{18} bits, so for $n = 8, 2^{15}$ n-tuples are taken.

Conceptually, each n-tuple addresses a RAM element. Although in the example above 2^{15} RAM elements, each of 256 bits, are needed these may be implemented using much fewer larger RAMs. Each array

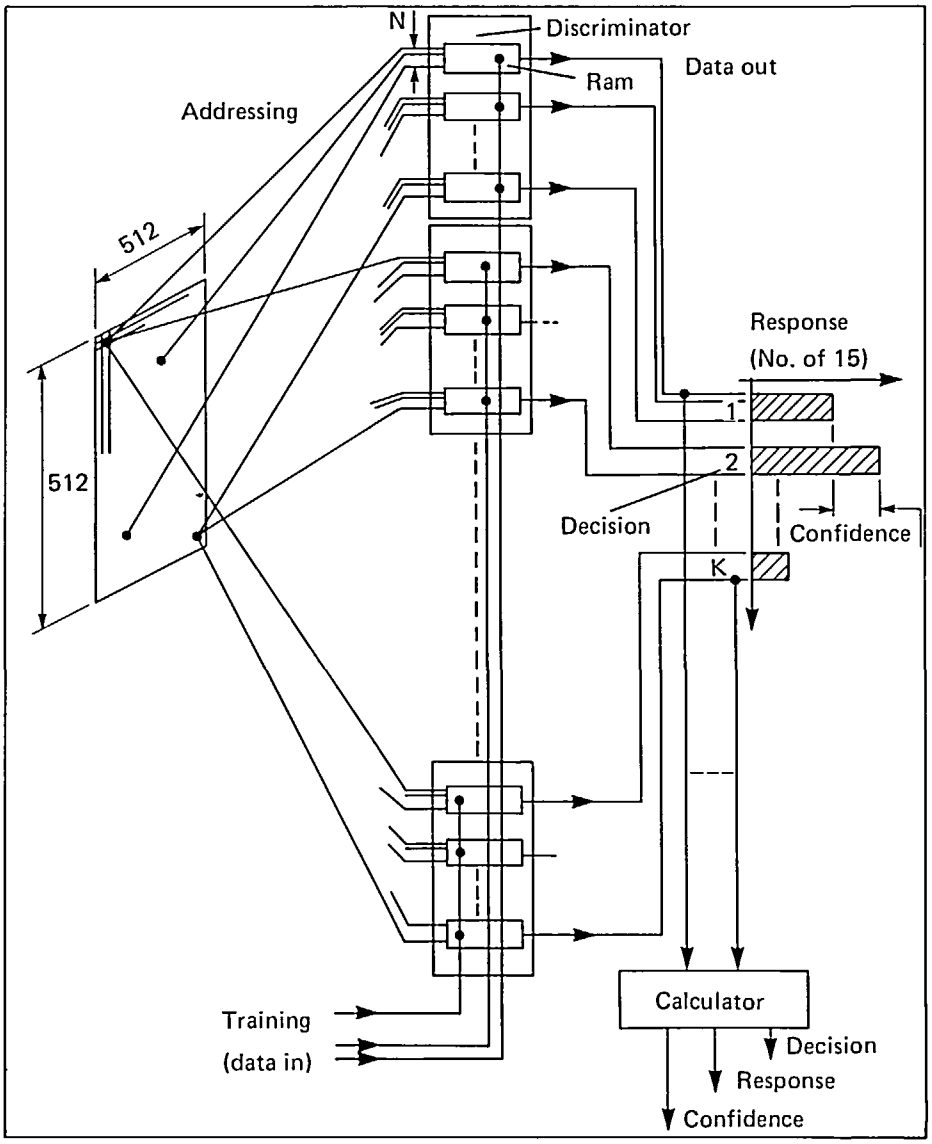


Fig 1. Notional structure of WISARD.

of such RAMs is called a discriminator. By using RAM arrays organised as K-bit words, K discriminators may be provided simultaneously.

The desired recognition activity is built up by a process of training on examples of the objects to be recognised. An object is presented through a television camera together with its desired classification. The classification selects a particular discriminator to be trained and, assuming that all the stores in that discriminator are set to zero to start with, ones are entered in all the locations addressed by that input image but only within the

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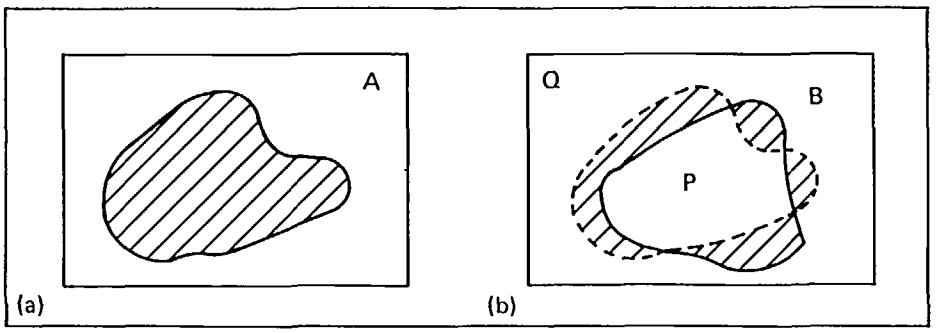


Fig. 2. Images A *and* B *have overlap area* Q+P.

selected discriminator. This has the effect of causing that particular discriminator to produce a logical one at all its data output terminals if the training image is presented again. If the input image is slightly changed not all the RAMs within the selected discriminator will respond with a one. The number of RAMs responding with a one quoted as a precentage of the total number of RAMs in a discriminator is called the response of that discriminator. These responses are fed into a simple calculator which first identifies which of the discriminators has the strongest response and outputs the class number associated with that discriminator. It also provides the actual response of the discriminator as well as the difference between the response of the strongest discriminator and the next one in line. This difference provides a measure of confidence with which the decision is made.

The full theory of this arrangement may be found elsewhere.[1,2] Here we provide a simple design principle from which the performance of the system may be estimated and understood.

Fig. 2(a) shows an object A (which is assumed to be binary for simplicity). A discriminator is trained on this image and tested with a second shape B, the outline of which is shown in Fig. 2(b). In order for a specific RAM to output a one, all its n address terminals must be found in the overlap area Q + P. If the total area of the image is D then it is easily shown that the probability of this event is:

$$[(Q + P)/D]^n$$

or, in general:

$$(\text{Overlap area}/\text{Total area})^n = L^n$$

This may be illustrated with an example involving a key. This is shown in Fig. 3. All the figures represent a key in a jig where Class 1 is represented by (a) and (b) where the key serrations are downwards, while in Class 2, (c) and (d) they are upward.

Two discriminators are used, one trained on Class 1 and the other on Class 2. Testing now with one of the training patterns, the correct discriminator will respond according to the white overlap area in (e) for which

$$L = 0.943$$

The training due to the second pattern is almost negligible in calculating this.

Confidence may be defined as:

$$C = (R_c - R_I)/R_c$$

where

R_c = response of correct discriminator.

R_I = response of nearer incorrect discriminator.

The response of the correct discriminator in this case is 100%, hence for this case:

$$C = 1 - L^n$$

and C may be tabulated against n

n =	1	2	4	8	16
%C=	5.7	11	21	37	61

Clearly this problem may be solved by template matching. This is equivalent in confidence terms to $n = 1$, that is 5.7%. The tabulation shows how much better figures may be obtained with the WISARD scheme and higher values of n. Against this, the demands on memory go up as 2^n, hence the maximum value of n is set by the availability of memory.

The importance of the confidence figure calculated above is due to the fact that it provides an indication of the degradation (say through noise) that is tolerated by the system before a misclassification occurs. The higher the confidence, the less likely is the misclassification. Using the above simplified theory, it may be shown that if 40% noise is present in the images of the keys, the following confidence levels may be achieved:

n =	1	2	4	8	16
%C=	1.6	2.7	7.7	11.8	18.6

This clearly shows how higher levels of n retain a workable performance in the face of considerable noise.

The above analysis has been presented for a simple training set only. Large training sets complicate

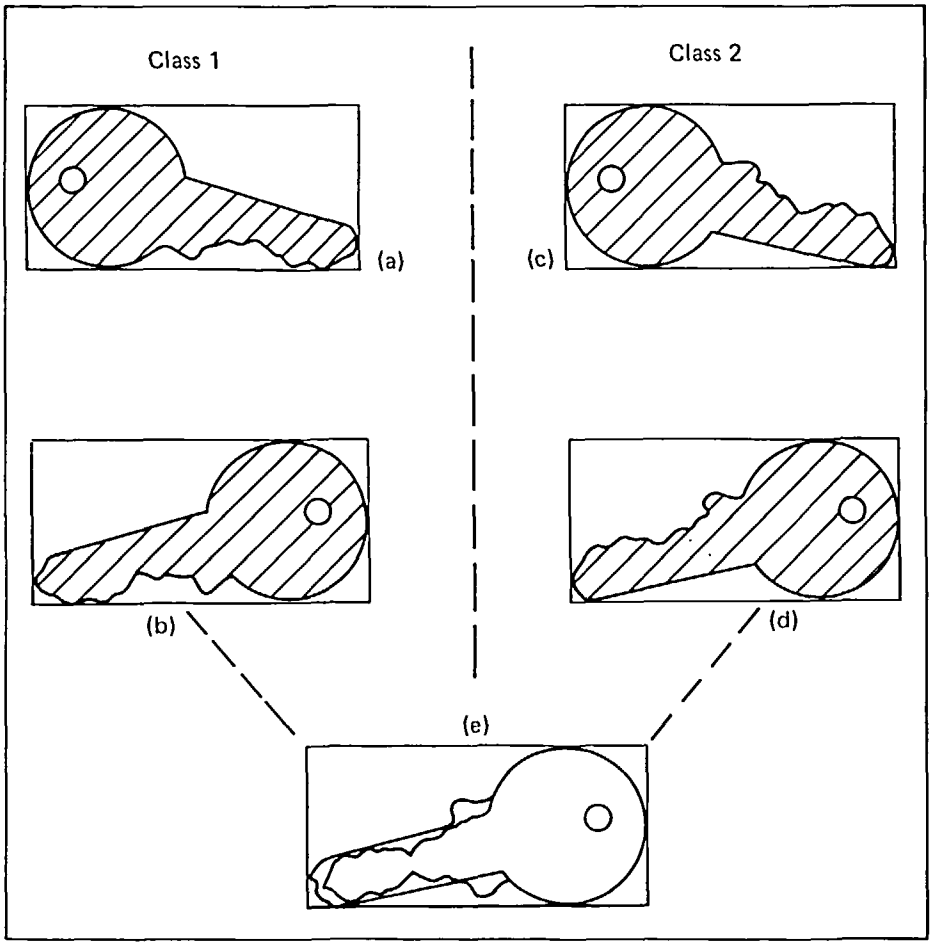


Fig. 3. Distinguishing key patterns.

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the analysis without altering the principle that n is available in the WISARD system which allows the user to set the discriminatory power of the system to the demands of a given problem.

Typically, the main advantageous properties of this type of system may be summarised as follows:

- A diversity of images (in terms of their overlap similarly) can be accepted by the training set. This has been demonstrated by using the system to recognise faces where, with a value of $n=8$, the training was achieved by asking a subject to move his head around in the field of view of the camera long enough for the system to take about 200 shots. This was repeated with 15 subjects who on subsequent presentations were recognised without error and with second responses between 20 and 30% down on the selected ones. The system acts as an excellent interpolator simply because an unknown image placed between two images seen during training is unlikely to map into a totally different image. Some of our theoretical work has demonstrated this effect in practical situations where mathematical models of simple images were not available.

  While pointing out the advantages of interpolation it is also important to point out the characteristics that are not available to the system. Good as the scheme is for interpolation, it is bad at extrapolation. This includes the inability to recognise an object rotated through say 180° where the system was only trained on the object rotating through 90°. In terms of faces this means that the system would not be able to recognise a profile having seen a front view. It is most unlikely that a human being could achieve this performance efficiently.
- Noise immunity is an important property of such systems. This is particularly true in cases where it is possible to train the system under non-noisy conditions. For example, an experiment has been carried out with intruder recognition in a situation where the system was trained on a scene not containing the intruder and under reasonable lighting conditions. Only one discriminator was used (that is one cover of the input image, the input image being an automatically thresholded 512 × 512 matrix of single bit pixels). It has shown that the intruder could be recognised through a jump in the response of the discriminator even if under testing conditions the lighting was reduced so as to have as much as 60% camera noise in the image.

  During training however, noise can be the factor which brings about the saturation and therefore total lack of performance of the system. Should applications arise where it is impossible to generate noise-free training data, such data may be pre-processed off-line. Of course, this does not mean that the system cannot cope with noisy training patterns, it simply means that if long exposures are required during training, the confidence with which subsequent decisions can be made will be reduced as a result of presence of noise in the training set.

Architecture

CRS has developed WISARD by adapting its proven, commercially successful, CRS1000 automatic inspection system. The CRS1000 has several attributes that make it suitable for this kind of development:

- Standard 625 line CCIR specification video input. Linescan input available as an option. Square pixel 14.75MHz sampling.
- Up to 16 full 8-bit deep 512 by 512 frame store with 6 or 8 bit A/D options.
- Contains a VMEbus computer with the image memory and discriminator memory being mapped directly into VME address space. VMEbus has a 16 Mbyte addressing range, so a large amount of image memory and discriminator memory can be supported.
- The image bus can be accessed by special purpose hardware modules as well as VMEbus modules.
- The image memory card can also be used as discriminator memory. 64K or 256K dynamic RAMs can be used enabling large discriminator memories to be created with minimum cost and space.
- There are several vacant slots in the VMEbus section, allowing many add on features such as Winchester/floppy discs, magnetic tape, extra RS232 or parallel ports, extra ROM/RAM for program use. This flexibility is essential to meet the wide range of industrial and scientific applications.
- The CRS1000 comes in a 6U rack allowing it to be easily incorporated in standard 19in format cabinets or desks for factory installations, or other stand alone applications.
- Software development packages for 68000/VME are now widely available, including UNIX operating systems, PASCAL, C and assembler compilers (including cross compilers for DEC equipment).

The block diagram of the WISARD/CRS1000 system is shown in Fig. 4. The system has a two bus structure, enabling pixels to be fetched from the image memory at full video rate, while simulataneously writing/reading data from the discriminator memory. This allows a complete WISARD classification sequence to take place in well under one frame time (40ms).

The WISARD process is started under software control, after a frame of video has been captured. The WISARD controller fetches n-tuples from the image store, thresholding the data with a programmable comparator. The address in discriminator memory is made up from an auto-incrementing counter and the n-tuple value itself, as in Fig. 5, where ABCDE is the 5-tuple read from the image store. N-tuples ranging from 4 to 8-tuple are sufficient for most industrial applications. If we assume that the WISARD controller visits all pixels in a x by y window, then the memory requirements for each discriminator are given by:

$$\text{memory size} = x, y. 2^n/n \text{ bits}$$

where n is the n-tuple size.

Up to 16 discriminators are directly supported by having a 16 bit wide discriminator memory, but more can easily be accommodated using time division multiplexing.

Two addressing modes for the n-tuple scanning sequence are supported. A two-dimensional pseudorandom addressing module is used for applications where no previous knowledge of the scene is available. This provides a repeatable random scanning pattern within a rectangular window in the image. For those applications where the scene has particular areas of interest, a programmable look up table (LUT) of addresses is provided. This is another image memory card, configured as 262144 by 16 bits, which allows novel scanning patterns to be implemented. For example, the addressing could concentrate on the curved cutting edge of a machine tool and ignore the rest of the scene, since it has little relevance.

When a discriminator is to be trained, the relevant bit plane is cleared to zero, up to the limit



Fig. 5. Address make-up in discriminator memory.

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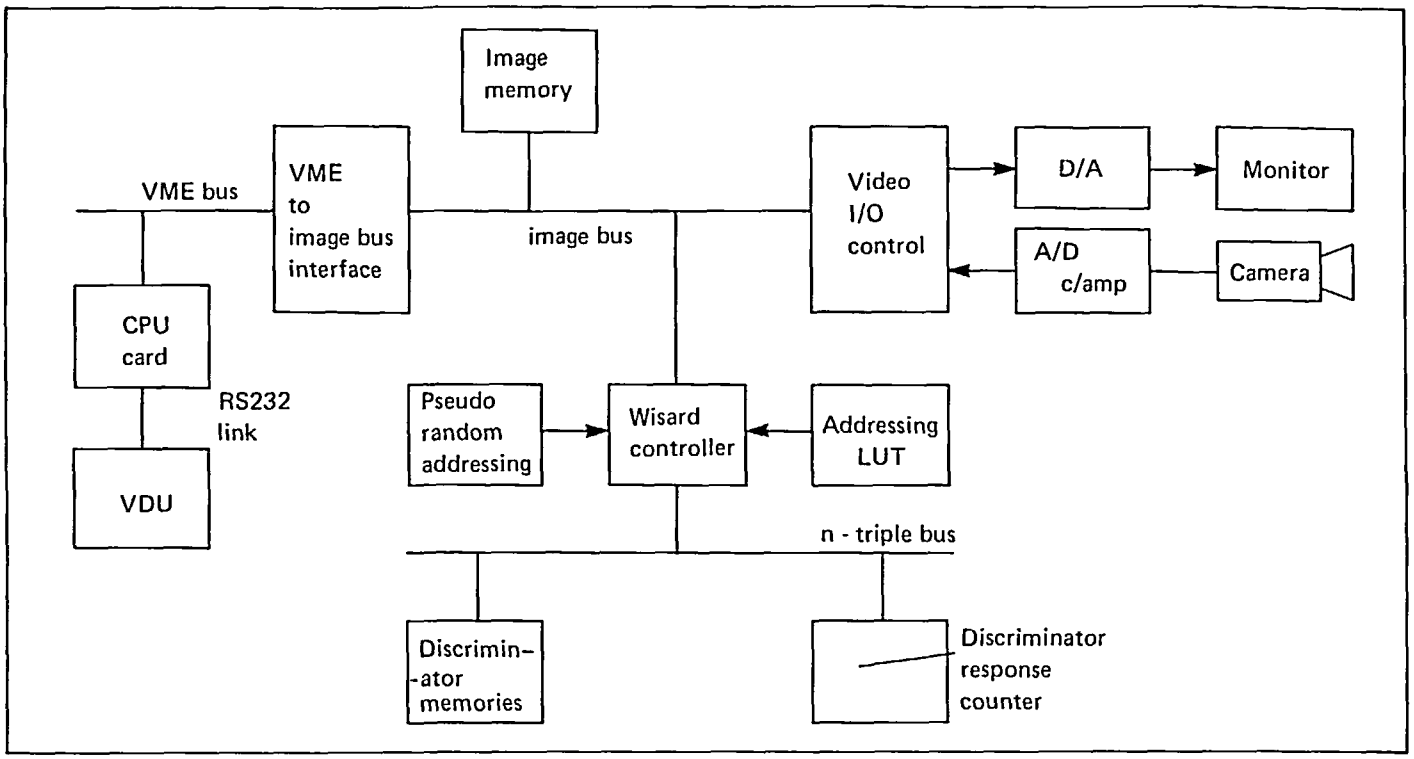


Fig. 4. Block diagram of WISARD and CRS1000 system.

imposed by n and the window size. A training scene is presented to the camera, the threshold is selected, and an image is captured. The pseudo-random addressing sequence commences, fetching n-tuples from the image.

For each n-tuple pattern that occurs a single 'one' is set in the appropriate 2^n segment of discriminator memory. The training sequence is repeated for all desired views of the object. During classification, rather than writing to discriminator memory, the locations accessed are read and a count is kept of all ones that occur. This count is read by the CPU at the end of the WISARD process and used to calculate the discriminator response and the confidence measure.

Scene classification is then simply a matter of choosing the maximum and next-to-maximum outputs from these discriminator response counters. Software can then activate 'failure' LEDs or relays, depending on the application.

The WISARD process is ideally suited to rapid go/no go classifications for production uses. Where actual measurement is also required, the direct access from VMEbus to image store makes this a straightforward task. The direct access from VMEbus to all aspects of the hardware make diagnostics simple, which is an essential part of equipment that must work reliably in an industrial environment. WISARD can be supplied with disk storage for backing up the discriminator memory, or can be trained using video tape recordings to minimise set-up times when the system is installed on a production line.

Applications

As a general rule, WISARD is most suitable for applications involving classification, solving problems in:

- Inspection and quality assurance – sorting good parts from bad.
- Identification and sorting – labels, shapes patterns and features.
- Security and surveillance –identification and movement or change detection.
- Robot vision – object location and robot guidance.

As an example, consider the inspection of a moulded or cast part. By varying the training images the inspection tolerances can be set. Training on a single view will make the inspection process critical. The smallest defect or distortion will cause a reduction in the discriminator output. By setting a threshold on this output, reject parts may be identified. However, it is often the case that certain features of the part have an allowable tolerance, whereas other features must be accurately positioned. This may be accommodated by varying the training set to include the extremes of allowable tolerance for each feature. Only when a feature is distorted beyond the allowable tolerance will the discriminator output drop. An important advantage is that training is a relatively unskilled process using sample parts. No part-specific programming is necessary.

Inspection of a single part needs only one discriminator. Using several discriminators, each trained on a different object, allows WISARD to identify and sort objects as they appear in the field of view. As in the earlier example, the keys, very similar objects can be distinguished. The generality of the objects can be accounted for during training. For example, the system can distinguish between 4mm bolts and 5mm bolts by training one discriminator on each size. If however, it is to distinguish between nuts and bolts of any size then one discriminator must be trained on bolts and one on nuts. The sensitivity and judgement ability of the system is controlled by the training process.

A specific case of sorting is label reading. Conventional character recognition techniques are able to read printed text on labels. WISARD can be trained to read text, but may be used more flexibly to recognise entire labels. It can cope with pictorial or symbolic information, warning signs or logos. In the drug industry it is vitally important that a packaged drug is correctly labelled. WISARD can ensure this with total reliability.

One interesting characteristic is its ability to detect changes in an image, even if the image is noisy. The intruder experiment described earlier is a real example. Although movement and change detectors for security applications are commercially available they do not function reliably unless the image quality is good. As image noise builds up, the WISARD discriminator output changes slowly but if the image itself suddenly changes then the discriminator output will jump. This

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unique capability arises because of the statistical properties of the process.

WISARD has been demonstrated as a device for verifying signatures.

Each of a set of discriminators is trained on specific signature features, such as vertical strokes, horizontal lines, loops and so on. WISARD then acts as a feature detector and signatures can be identified by their characteristic features. This approach can be used for other applications where many different objects need to be identified.

The system is not primarily intended for applications where measurements are necessary. However, it can be used to locate objects and assess their position or orientation. Position may be gauged in two ways. The first method uses a scanned window approach and is similar to correlation. A small window within the image, containing the object, is used to train a discriminator. Scanning the window over the image and finding the point of maximum response from the discriminator allows the object to be accurately located. This technique has the disadvantage that 40ms is taken for each position of the window so to locate an object precisely quite a long time may be needed.

However, using the second approach, an object may be located much more quickly. WISARD is very good at interpolation, so four discriminators looking at the full field of view are all that is required to locate an object in one frame time of 40ms. One discriminator is trained with the object in various positions at the top of the field of view, one with the object at the bottom, one at the left and one at the right. The position of the object in the field of view may be determined by examining the relative outputs of the four discriminators. This same property of interpolation may be used to gauge rotation or orientation using only a small number of discriminators.

Future directions

In most of the cases described above, the classification process is performed in one television frame time of 40ms. Some applications require several stages of classification, but rarely will processing time be a limitation using the technique.

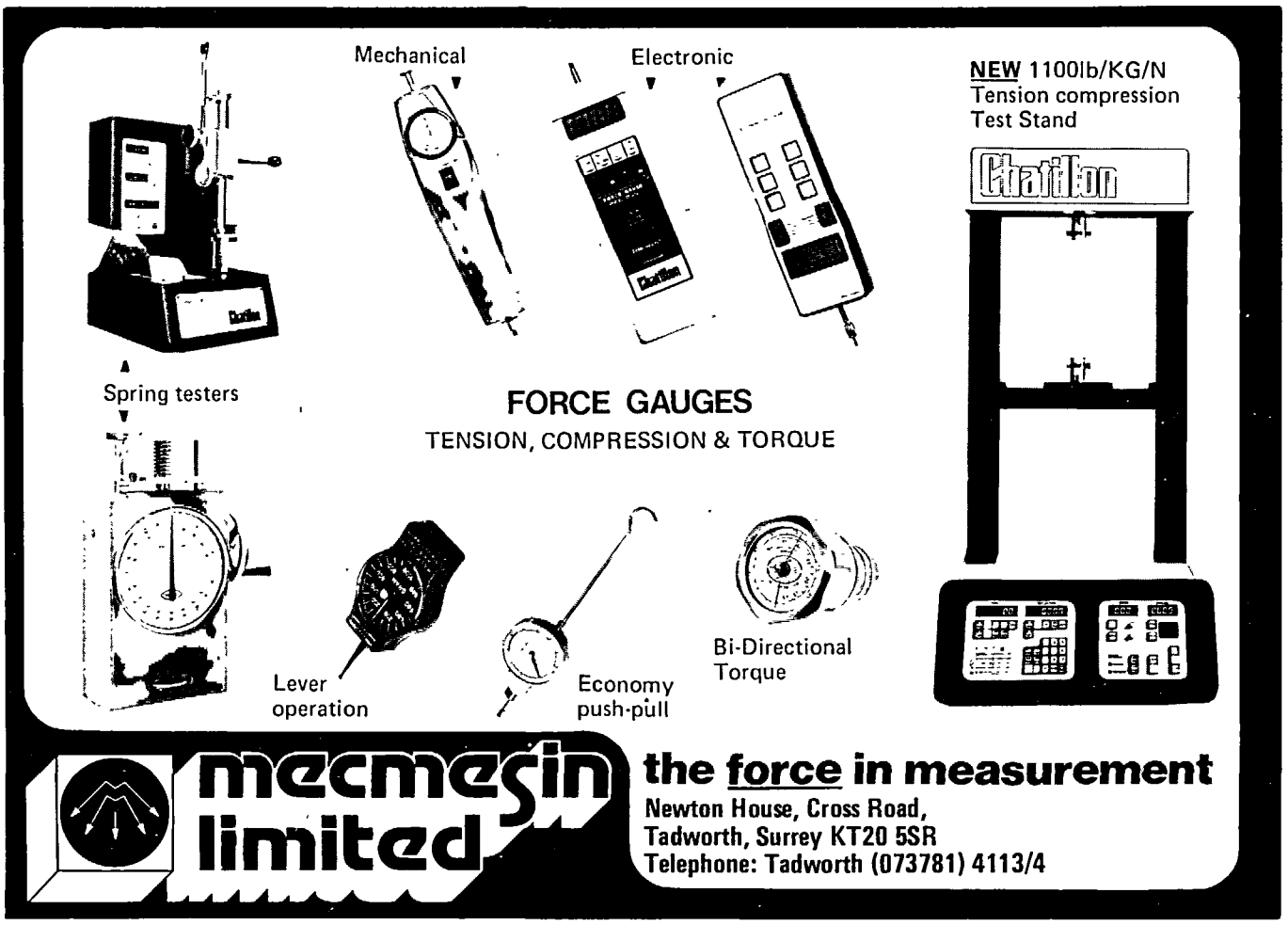
Research continues at Brunel and at CRS to find new ways of using the technique to increase still further its applicability and effectiveness. One example is the use of feedback. Feedback gives WISARD new properties including the ability to identify sequences of images and to associate an image with a prototype or concept.[3] For example, given the image of a bolt, WISARD itself can recall the image of an ideal bolt.

An interesting adaptation of the technique allows objects to be tracked in a way similar to the human visual system. A discriminator network which covers the whole image, but has a steerable central higher-resolution region is used. Positional feedback is applied to track the object rather like the way the foveal region of the retina centres on objects of interest. With several objects or features in the image, the high resolution region will jump from one to another

The WISARD system will be on view together with many other image processing products on the CRS stand at the Machine Intelligence Exhibition to be held in London from 9-11 October 1984.

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